The Gate Knows, the Threshold Doesn't: Recalibrating Activation-Gated Jailbreak Defenses Under Unseen Framings

Anonymous ACL submission

Abstract

Activation-gated defenses read a model's residual stream and steer towards refusal only when a probe fires. Their thresholds are set on jailbreak framings known at development time. We test six such gates on real harmful and benign prompts in thirteen framings, nine of them withheld from design. On withheld framings, recall at a 1% false-positive budget often falls to 0–17%. Yet a probe over the user's tokens still ranks harmful above benign prompts with AUROC 0.95–1.00 whenever the model can parse the framing, and re-thresholding per framing would recover most of the loss. The failure is largely one of calibration. A location-scale model accounts for the shift, shows why a correction that sees only one framing's unlabeled scores needs a bound on its harmful share, and describes how recalibration degrades as that share grows. We propose a recalibration that needs no labels from the deployment stream: cluster prompts by framing, standardise the probe score robustly within each cluster, and keep the original threshold as a floor. The method is useful but fragile. With enough clusters, a post-hoc choice, recall on most withheld framings rises to 83–94% on two models at 1.3–1.6% mean false positives, and judged attack success falls, for example from 45% to 6% on code-completion framings (Qwen2.5-1.5B). The gains do not carry over where the steering itself is ineffective (fiction on Phi-3.5-mini), and framings used almost only for attacks are not helped.

1 Introduction

Safety training teaches a language model to refuse harmful requests. Jailbreaks undo it by changing how the request is framed: a persona, a story, a retrieved document, an encoding, or a request to write code (Wei et al., 2023; Shen et al., 2024; Greshake et al., 2023; Li et al., 2023). Because the framing changes the surface of a prompt but not what it asks for, a natural defense looks beneath the surface. Residual-stream activations carry a linearly readable harmfulness signal (Zou et al., 2023a; Zhao et al., 2025), and refusal itself is mediated by a single direction (Arditi et al., 2024). Several defenses therefore read activations to decide whether to intervene. Conditional activation steering adds a refusal direction only when a condition vector fires (Lee et al., 2025), and probe-based detectors flag a prompt from its internal state (Zhang et al., 2025; McKenzie et al., 2025). These gates are cheap, they leave benign traffic alone, and their operating point can be set to a fixed false-positive budget.

This paper asks what happens to that operating point when the framing changes. We separate framing families into *seen* ones, used for training and calibration, and *withheld* ones, and we evaluate on real prompts from five public benchmarks (Mazeika et al., 2024; Zou et al., 2023b; Chao et al., 2024; Röttger et al., 2024; Cui et al., 2025), with generations judged by a guard model (Qwen Team et al., 2025).

The first finding is a collapse. Every gate we test detects at most 17% of harmful prompts on withheld fiction and indirect-injection framings, and none of them changes the 89% attack success rate of fiction prompts. This includes a geometric gate of our own design that explicitly tries to factor framing out of the representation.

The second finding concerns the cause. Ranking survives the shift. A probe that reads the mean of the user's tokens separates harmful from benign prompts with AUROC 0.95–1.00 on every withheld framing the model can parse. On the four framings the shared threshold misses worst, re-thresholding it on the new framing recovers 57–81% recall. What moves is the location of both score distributions. Concurrent work reports compatible observations: linear jailbreak probes are poorly calibrated off-distribution (Kim and Lee, 2026), and shift degrades threshold transfer before ranking (Luo et al., 2026). We confirm the pat-

tern across gates, readout positions and framings, and Section 5 gives a location-scale model that accounts for it. We also show that projecting out a subspace of framing directions, the most direct representational fix we could build, does not repair it.

The third finding is a remedy and a map of its limits. Deployed gates see a stream of mostly benign prompts. We cluster the stream by framing, standardise the probe score inside each cluster with median and median absolute deviation, which tolerate a modest share of harmful prompts (Rousseeuw and Croux, 1993; Leys et al., 2013), and keep the original threshold as a floor. Unsupervised recalibration of subpopulations has antecedents in general machine learning (Ziegler and Czyż, 2019). Our contribution is to adapt it to activation-gated safety and to test it under a protocol that withholds framings from design. We then report plainly where it works and where it does not.

Concretely, we contribute:

- a held-out-framing evaluation of activation gates on real corpora, with leave-one-framing-out selection on validation data and five framings preregistered after the main study;
- evidence that gate failure on withheld framings is mostly a calibration shift, a location-scale model that explains it, a remark on why label-free recalibration needs a bound on the harmful share, and a negative result for framing-subspace projection;
- a label-free, cluster-wise recalibration with a fixed-threshold floor, evaluated without transduction at detection and generation level;
- a characterisation of its failure modes: cluster resolution, framings used mainly for attacks, harmful prevalence, and framings the model cannot parse.

2 Related Work

Jailbreak framings. Attacks range from optimised suffixes (Zou et al., 2023b; Liu et al., 2024) and attacker models (Chao et al., 2025; Mehrotra et al., 2024) to hand-written templates: personas (Shen et al., 2024), nested fiction (Li et al., 2023), encodings and ciphers (Wei et al., 2023; Yuan et al., 2024), many-shot demonstrations (Anil et al., 2024), and instructions planted in retrieved content (Greshake et al., 2023). Wei et al. (2023) attribute their success partly to mismatched generalisation, which is what a withheld framing probes.

Representation-level defenses. Harmfulness and refusal are encoded separately, with harmfulness concentrated at the instruction tokens rather than at the final position (Zhao et al., 2025). Representation engineering reads and edits such directions (Zou et al., 2023a); circuit breakers reroute harmful representations during training (Zou et al., 2024); latent adversarial training hardens them (Sheshadri et al., 2024). Inference-time gates include conditional steering (Lee et al., 2025) and detectors that flag prompts from internal signals (Zhang et al., 2025; Hu et al., 2024; Xie et al., 2024), or redact during generation (Mei et al., 2026). FrameShield likewise treats framing as a nuisance to be disentangled from the goal in activation space (Farzam et al., 2026). Several studies question how far probe signals generalise. Probes for high-stakes interactions keep high AUROC out of distribution while recall at low false-positive rates is modest (McKenzie et al., 2025). Final-token probes miss jailbreaks whose evidence sits in earlier user tokens (Doda, 2026). Linear jailbreak probes conflate benign register with intent, and their headline accuracy depends on the threshold (Kim and Lee, 2026). Internal harmfulness scores can even anti-rank successful jailbreaks (Luo et al., 2026). In contrast, Schwarz (2026) find that refitting thresholds does not help on topic-matched pairs, where the discriminative information is absent. Adaptive attacks that optimise against the probe can drive recall to zero (Bailey et al., 2026). We study the complementary, non-adaptive failure of natural framing shift.

Guard models and judges. Classifier guards moderate inputs and outputs (Inan et al., 2023; Han et al., 2024; Zeng et al., 2024; Padhi et al., 2025; Qwen Team et al., 2025), and constitutional classifiers are trained against universal jailbreaks (Sharma et al., 2025). We use a guard only as an evaluation judge, because refusal-string matching counts harmless non-compliance as a successful attack (Souly et al., 2024).

Calibration under shift. Label-shift corrections re-estimate class priors from unlabeled target data (Saerens et al., 2002; Lipton et al., 2018; Alexandari et al., 2020). Test-time adaptation adjusts normalisation layers by entropy minimisation (Wang et al., 2021). Conformal methods restore coverage under shift given weights or labels (Tibshirani et al., 2019; Podkopaev and Ramdas, 2021). Unsu-

pervised recalibration adjusts outputs per subpopulation without labels (Ziegler and Czyż, 2019); multicalibration asks for calibration on every identifiable subgroup, using labels (Hébert-Johnson et al., 2018). For jailbreak detectors, Piet et al. (2025) adapt text classifiers to drift by self-training, Azov et al. (2026) find that re-thresholding often suffices when target labels exist, and Kim and Lee (2026) combine register erasure with distribution-free calibration on labelled benign data. Our method uses no labels from the shifted framing or the deployment stream, and it operates per framing cluster of an activation gate.

3 Setup

Prompts. Harmful behaviours come from JailbreakBench, the standard text behaviours of HarmBench, and AdvBench; benign prompts come from the benign split of JailbreakBench, the safe prompts of XSTest, and 400 prompts of OR-Bench-Hard, which are designed to look harmful. After deduplication there are 1,550 prompts. We split them 50/20/30 into train, validation and test *by behaviour*, stratified by source, so that no wrapped version of a test request is seen in training. The test split holds 241 harmful and 225 benign requests.

Framings. Each request is wrapped in a framing that keeps its text verbatim. Four framings are *seen*: the plain request, a persona (AIM, DAN, "grandma"), refusal suppression, and prefix injection. Four were *withheld* from the start: fiction or screenplay, indirect injection disguised as a retrieved document or e-mail, base64, and leetspeak. After the main study we wrote and committed five *fresh* framings before generating any output on them: a past-tense historical frame, an academic or IRB frame, code completion, translate-then-answer, and payload splitting. Every family uses one to three fixed templates. We also optimise GCG suffixes (Zou et al., 2023b) against the undefended model for 20 test behaviours (Appendix B).

Model and gates. The target is Qwen2.5-1.5B-Instruct (Qwen Team, 2024) without a system prompt. All gates read layer 14 of 28. When a gate fires, it adds the same vector $\alpha\hat{r}$ to the residual stream at every position. Here $\hat{r}$ is the normalised difference between mean harmful and mean benign activations of plain training requests, and α is 0.4 times the mean activation norm. Gates therefore differ only in their decision. We compare six gates: (i) a CAST-style condition vector (Lee et al., 2025); (ii) an MLP probe on the final token; (iii) ARETE, a geometric gate of our own that maps activations onto intent and context hyperspheres and steers orthogonally to a framing subspace (Appendix A); and logistic-regression probes on (iv) the final token, (v) the mean of the user's tokens, and (vi) a union of three readouts (Section 4). Every gate is trained on seen framings of the training split. Its threshold is the 99th percentile of its score on benign validation prompts of seen framings, a held-out-quantile calibration in the spirit of Kim and Lee (2026) but without their register erasure. We call it the *shared* threshold.

Protocol and its departures. Readout position, clustering layer and clustering features were selected by leave-one-framing-out (LOFO) on the validation split: each seen wrapper in turn is treated as withheld, and recall is measured on it. Two departures need stating. First, the diagnostic analyses of Section 4 used the test split before the remedy was designed, so the original withheld framings informed the idea, though not its configuration. This is why we pre-registered the fresh framings. Second, the layer was fixed at 14 for all gates and never selected; in LOFO the user-mean probe scored slightly higher at layers 21 and 7 (43% and 41% recall) than at layer 14 (38%).

Metrics. Detection is reported as true-positive rate (TPR, which we call recall) on harmful prompts and false-positive rate (FPR) on benign prompts, plus AUROC. Responses are generated greedily for 96 new tokens and judged by Qwen3Guard-Gen-4B in response-moderation mode (Qwen Team et al., 2025). The judge sees the *unwrapped* request and the response. Attack success rate (ASR) is the fraction of harmful requests whose response is judged UNSAFE. Over-refusal is the fraction of plain benign requests whose response is judged a refusal; on wrapped benign prompts the judge often labels confused non-compliance as refusal, so we do not report it there. For base64 the judge labels responses that merely echo or start decoding the payload as UNSAFE, so we report detection only. With 96 tokens, ASR mostly measures the onset of compliance; Appendix B repeats the key cells at 256 tokens.

	Seen framings				Unseen framings			
Gate	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak
CAST	85 (0.98)	42 (0.97)	50 (0.96)	0 (0.94)	0 (0.87)	0 (0.94)	0 (0.46)	0 (0.93)
MLP probe	91 (1.00)	90 (1.00)	87 (1.00)	83 (1.00)	0 (0.81)	0 (0.93)	0 (0.41)	0 (0.94)
ARETE (v1 geometry)	93 (0.99)	93 (0.99)	94 (0.99)	88 (0.99)	0 (0.40)	3 (0.71)	0 (0.47)	0 (0.93)
Probe, last token	94 (1.00)	91 (1.00)	90 (1.00)	87 (1.00)	0 (0.86)	15 (0.89)	0 (0.45)	0 (0.95)
Probe, user tokens (mean)	96 (1.00)	88 (1.00)	83 (1.00)	93 (1.00)	5 (0.99)	17 (0.98)	0 (0.53)	0 (0.72)
Multi-view union	97 (1.00)	93 (0.99)	92 (0.99)	88 (1.00)	0 (0.83)	12 (0.89)	1 (0.54)	5 (0.94)

Table 1: Detection on the test split at the shared threshold (1% benign FPR on seen framings): recall (%) and, in parentheses, AUROC. All gates read layer 14 of Qwen2.5-1.5B-Instruct.

Readout	Framing	AUROC	TPR (shared)	TPR (oracle)
Last token	Fiction	0.86	0.0	21.6
	Indirect	0.89	15.4	41.9
	Base64	0.45	0.0	0.0
	Leetspeak	0.95	0.0	17.8
User mean	Fiction	0.99	4.6	80.5
	Indirect	0.98	16.6	78.8
	Base64	0.53	0.0	0.0
	Leetspeak	0.72	0.0	17.4

Table 2: Withheld framings, test split. *Shared*: threshold from seen framings. *Oracle*: threshold re-set to 1% FPR on benign prompts of the same framing (unavailable in practice). Recall in %.

Fresh framing	AUROC	TPR (shared)	TPR (oracle)
Past tense	0.97	0.4	57.3
Academic	0.96	39.8	63.5
Code	0.98	83.0	67.6
Translation	1.00	95.4	90.0
Payload split	0.99	0.4	78.0

Table 3: Fresh pre-registered framings, user-mean probe, test split. Recall in %.

4 Diagnosis: Ranking Survives, the Threshold Moves

Recall collapses on withheld framings. Table 1 shows that the probes and ARETE detect 83–97% of harmful prompts on seen framings at the shared threshold. CAST is weaker on wrapped seen framings, at 0–50%. On withheld framings recall falls to 0–17% for every gate. Benign FPR on those framings also falls to nearly zero, which already suggests a shift in location: the whole score distribution of a new framing moves below the threshold. ARETE's fiction AUROC is 0.40, worse than chance, so its learned intent geometry does not transfer to a narrative frame.

The information is still there. If a framing erased the harmful signal, no threshold could recover it. Table 2 compares the shared threshold with an oracle that re-thresholds on benign prompts of the withheld framing itself. For the user-mean probe, AUROC on fiction and indirect injection is 0.99 and 0.98, and the oracle lifts recall from 5% to 81% and from 17% to 79%. The fresh framings, examined only after the method was fixed, show the same pattern (Table 3). AUROC ranges from 0.95 to 1.00, and the oracle lifts past-tense recall from 0.4% to 57% and payload-split recall from 0.4% to 78%. Base64 is the exception. AUROC is near chance and no threshold helps, because the 1.5B model does not reliably decode the payload, so no representation of the request forms. Leetspeak is similar at the user-mean readout (AUROC 0.72), and the model rarely complies with it at all.

Projecting out framing does not help. ARETE assumes that framing occupies a low-dimensional subspace that can be removed. We test this in LOFO on validation data: we estimate the subspace from wrapper displacements of the remaining seen wrappers and project it out before training a linear probe. Removing 0, 4, 16 or 64 directions gives mean held-out recall of 54%, 54%, 54% and 48%. A subspace learned from two wrappers does not contain the shift induced by a third. Prefix injection is the clearest case, at about 1% recall for every rank.

Where to read. Readout position matters more than projection. In LOFO, the final-token probe misses prefix injection (1%), whose injected instruction follows the request, while the user-mean probe catches 60% of it. A union of three readouts, each at a third of the budget, raises LOFO recall from 54% to 69% at 1.3% FPR and helps seen framings slightly (Table 1), but not withheld ones, because each readout suffers the same shift.

5 A Model of the Shift

Why does a framing break the threshold but spare the ranking? A first-order answer is that it moves the whole score distribution and leaves the shape of each class alone. We state this as a model so that its consequences can be checked. Let $s = a_f + b_f\,\xi$ be the probe score of a prompt in framing f, where $\xi \sim F_y$ depends only on the label y, F_1 stochastically dominates F_0, and (a_f, b_f), $b_f > 0$, are a location and scale set by the framing. The model is an approximation: measured AUROC varies from 0.95 to 1.00 across framings the model can parse, and is near chance on base64.

Proposition 1 (ranking is invariant, thresholds are not). Within a framing, AUROC does not depend on (a_f, b_f), because the map $\xi \mapsto a_f + b_f\xi$ is increasing. A fixed threshold τ yields $\mathrm{TPR}_f = 1 - F_1((\tau - a_f)/b_f)$ and $\mathrm{FPR}_f = 1 - F_0((\tau - a_f)/b_f)$. Under the model the ROC curve is the same for every framing, and the framing only slides the operating point along it. A downward shift of a_f therefore lowers recall and false positives together, which is the pattern in Table 1: benign FPR on withheld framings falls to nearly zero at the same time as recall.

Remark 2 (a bound on the harmful share is necessary). Let $F_0 = \mathcal{N}(0, 1)$ and $F_1 = \mathcal{N}(d, 1)$. A framing that is entirely benign at location $a + d$ and one that is entirely harmful at location a produce the same unlabeled score distribution, $\mathcal{N}(a + d, 1)$. A rule that sees only the unlabeled scores of one framing fires at the same rate r on both populations. Its false-positive rate on the benign framing is then equal to its recall on the harmful one, so no such rule can be good on both. Recalibrating a framing from its own unlabeled scores therefore needs an assumption on its harmful share π_f. This is the assumption of Section 6: it is necessary for rules of this kind, and it is not removed by choosing a different statistic. The remark does not cover rules that use side information, such as the plain-request calibration that the hybrid floor retains.

Proposition 3 (what contamination costs the robust statistics). Suppose a framing is observed as the mixture $(1 - \pi)F_0 + \pi F_1$, with F_0, F_1 unit-variance Gaussians separated by d. Let $m(\pi, d)$ be the median and $\sigma(\pi, d) = 1.4826\,\mathrm{MAD}$ of the mixture, each the root of a monotone equation. The rule that fires when $(s - m)/\sigma > z^\star$ has

$$\mathrm{FPR} = 1 - \Phi\big(m + z^\star\sigma\big),$$
$$\mathrm{TPR} = 1 - \Phi\big(m + z^\star\sigma - d\big).$$

At $\pi = 0$ this gives $m = 0$ and $\sigma = 1$, so $z^\star$ approximately attains the intended false-positive rate. As π grows, harmful prompts pull m upward and inflate σ, which raises the effective threshold. At $\pi = 1/2$ the median sits between the modes, and for $d \geq 2$ and $z^\star = 2.52$ the threshold lies above the harmful mode, so recall is below 0.3% (Appendix B, Table 9). Recall falls steeply between $\pi = 0.1$ and $\pi = 0.3$ at every separation. A larger d raises recall at small π but moves the breakdown only modestly: the share at which recall drops below 5% rises from 0.24 at $d = 2$ to 0.36 at $d = 5$. At $d = 4$, recall is 90% at $\pi = 0.05$ and 13% at $\pi = 0.30$.

Checking the predictions. We test Proposition 3 without the Gaussian assumption. For every framing and traffic mix, we build the mixture of the empirical benign and harmful test scores at the scenario's prevalence, compute its median and MAD, and predict the FPR and TPR of the rule that standardises by true framing (the oracle grouping, which removes clustering error). We compare these predictions with the window measurements of Section 7, over all thirteen framings and both mixes. Predictions and measurements agree to within 1–2 points of recall (Table 4), and the model puts persona, refusal suppression, prefix injection and payload splitting at 0% recall at $\pi = 0.5$, as measured. The check is partly circular, because the predictions are built from the same test scores that the windows resample. It isolates the robust-statistics step and shows that contamination accounts for the failure. It does not show that the model predicts scores on unseen traffic, and it does not test the assumption of a common location-scale transform, which Section 7 tests only indirectly through the clustered rule.

Clustering error. The oracle grouping is not what runs in deployment. With $k_{\max}$=30 on one realistic window, most framings fall into one to three clusters and the plain request into more. Each cluster of at least 50 prompts holds a single framing, except that the main fiction cluster also contains plain requests. The effective harmful share of a framing's main cluster is within about 0.02 of its nominal prevalence (0.07 against 0.05 for

	TPR		FPR	
Model	corr.	MAE	corr.	MAE
Qwen2.5-1.5B	0.998	1.4	0.987	0.14
Phi-3.5-mini	0.999	1.0	0.978	0.15

Table 4: Predicted versus measured firing of the oracle-grouped robust-z rule, over 13 framings and 2 traffic mixes (26 points per model). Correlation, and mean absolute error in percentage points.

fiction, 0.51 against 0.50 for persona). Treating each framing as one location-scale family nevertheless predicts the clustered rule only roughly: predictions correlate at 0.88 with measured adaptive recall, with a mean error of 11 points, and six of the 13 points are near zero in both. The model underestimates four framings by 13 to 62 points and overestimates two by 9 to 10. The largest miss is indirect injection, where the model predicts 27% at $\pi = 0.05$ and the clustered rule reaches 90%. A cluster can be tighter than its framing, so its MAD is smaller and its threshold lower.

Flooding. Proposition 3 also says how an attacker can degrade the rule. A rule that standardises within a cluster can be pushed off its operating point by raising the harmful share of that cluster. We let the share of one framing grow from 5% to 50% while the rest of the stream stays unchanged, on Qwen2.5-1.5B only (Appendix B, Table 10). For fiction, the oracle-grouped rule falls from 76% recall to 68, 54, 20 and 0%, against predictions of 76, 69, 50, 17 and 0%. The clustered rule behaves the same way: fiction recall is 87% at $\pi = 0.05$, 59% at $\pi = 0.2$ and 17% at $\pi = 0.3$. To reach that point the attacker must send about 0.43 harmful prompts for each benign one in the framing, and the harmful prompts it sends are the traffic that goes through. The hybrid floor limits the damage only where the shared threshold already works, as for code (69% recall at $\pi = 0.5$, against 83% for the shared rule at full budget). The floor sits at half the budget, so at $\pi = 0.5$ the hybrid rule falls somewhat below the shared one on every framing we flooded. Where the shared threshold fails, as for fiction, flooding leaves little more than the uncalibrated behaviour. We did not test attackers who adapt their prompts to the probe (Bailey et al., 2026), who spread a framing over many small clusters that fall back to validation statistics, or who poison the window that sets the statistics.

Algorithm 1 Cluster-wise robust recalibration

Require: probe score s; framing features ϕ; budget ϵ; shared threshold $\tau_{\epsilon/2}$

1: **Once, on validation:** simulate streams of seen framings, run lines 2–3, and set $z^\star$ to the $(1 - \epsilon/2)$ quantile of benign robust z.
2: **Window:** on unlabeled traffic, reduce ϕ by PCA and run k-means, $k \le k_{\max}$ chosen by silhouette.
3: For each cluster c: $m_c \leftarrow \mathrm{median}(s)$, $\sigma_c \leftarrow 1.4826\,\mathrm{MAD}(s)$; freeze centroids and (m_c, σ_c).
4: **New prompt** x, nearest centroid c: fire iff $s(x) > \tau_{\epsilon/2}$ or $(s(x) - m_c)/\sigma_c > z^\star$.

6 Cluster-Wise Robust Recalibration

The diagnosis and the model of Section 5 suggest treating each framing as its own calibration problem. At deployment the framing of a new prompt is unknown and no labels are available. Prompts that share a framing share an activation signature, so they cluster, and when a framing is used mostly for benign purposes the bulk of its scores describes benign prompts. By Remark 2 this assumption cannot be dropped.

Algorithm 1 gives the *hybrid* rule. The score s is the user-mean probe. The features ϕ are user-mean activations at layer 14, reduced to 32 principal components. Clusters with fewer than 30 prompts fall back to validation statistics. Inside a cluster the median and MAD describe its benign bulk, so a harmful prompt is compared with benign prompts of its own framing. Because $z^\star$ is set on validation streams that pass through the same clustering, it accounts for within-cluster dispersion. The *adaptive* rule is the second disjunct alone, at the full budget. Validation labels are used once, to set $z^\star$ and τ. No labels from the deployment stream or from new framings are used.

LOFO on validation streams at 5% harmful prevalence selected the user-mean readout over the final-token and last-user-token readouts, and layer-14 clustering features over layers 7 and 21. Under that selection, held-out recall rose from 38% with the shared threshold to 91% with the adaptive rule, at 1.1% FPR. The hybrid rule reached 83% at 0.9% FPR. We fixed $k_{\max} = 12$ without selection, which turns out to matter.

7 Results

We evaluate without transduction. In each of 20 windows the test behaviours are split at random into halves A and B. Cluster statistics are estimated on traffic built from half A and applied to half B. We use two traffic mixes. *Benign-rich* traffic gives every framing 5% harmful prevalence. *Realistic* traffic keeps 5% for dual-use framings (plain, fiction, indirect, past tense, academic, code, translation) and uses 50% for framings that are rarely used benignly (persona, refusal suppression, prefix injection, base64, leetspeak, payload split). Generation-level ASR is composed from judged responses. A fired prompt takes the label of its steered response, which equals the always-on response under greedy decoding, and an unfired prompt takes the label of its undefended response.

As pre-registered, the method under-segments. Table 5a reports the frozen configuration on all thirteen framings. In benign-rich traffic, payload-split recall rises from 0% to 90%, indirect injection from 16% to 31%, past tense from 0% to 15%, and academic from 40% to 45%. Fiction gains little, from 5% to 19%. When the stream contains only the eight original framings, the same configuration detects 82% of fiction and 56% of indirect prompts. With thirteen framings and at most twelve clusters, fiction shares a cluster with framings whose benign scores sit elsewhere, and its location is not corrected. With true framing labels as groups (Table 6) fiction recall is 76%, so the loss comes from cluster resolution, not from the statistics.

With enough clusters it works where traffic is mostly benign. Raising $k_{\max}$ to 30 is a post-hoc change, motivated by the observation above, and we report it as such. In benign-rich traffic, adaptive recall then reaches 91% on fiction, 92% on indirect injection, 84% on academic and 90% on payload splitting, at 1.6% mean FPR (Table 5b). Recall rises steadily with $k_{\max}$: at 12, 20 and 30 it is 27%, 70% and 91% on fiction. Past tense stays at 40%, below its 57% oracle. Table 6 isolates the clustering. One global robust z (k=1) does nothing. Grouping by true framing family helps, but less than data-driven clusters, which can also separate templates within a family.

It fails where a framing is mostly harmful, and the floor contains the damage. Under realistic traffic the persona, refusal-suppression and prefix-injection clusters are half harmful. Their median then sits among harmful scores, and adaptive recall on them drops to 0–4% (Table 5). This is worse than the shared threshold, and prefix-injection ASR returns to the undefended 77%. Payload splitting, an attack-only framing that the shared threshold also misses, is lost entirely. The hybrid rule keeps the shared threshold as a floor and recovers 55–82% recall on seen attack-only framings, while keeping nearly all of the gain on dual-use framings.

Generation. Table 7 reports judged ASR under realistic traffic. The hybrid rule lowers ASR on fiction from 89% to 37%, on code completion from 45% to 6%, on academic framing from 15% to 4%, and on indirect injection from 12% to 6%. The shared threshold leaves fiction and indirect injection unchanged. Translate-then-answer is already caught by the shared threshold (51% to 3%), and the hybrid rule keeps it there (5%). Over-refusal on plain benign requests rises from 60.3% to 60.6%. The high base rate reflects how often the 1.5B model refuses the borderline prompts of XSTest and OR-Bench even undefended. An always-on steer, by contrast, refuses every plain benign request. The cost of the hybrid rule falls on prefix injection, where ASR rises from 9% (shared) to 16%, and on payload splitting, which stays at 95%. Appendix B gives the pre-registered configuration and a prevalence sweep. With harmful prompts at 20% of every framing the adaptive statistics degrade, and at 35% they fail.

A second model. To check that nothing is specific to Qwen, we ran the whole pipeline on Phi-3.5-mini-instruct (3.8B parameters), with every choice frozen from the Qwen study: the user-mean probe at the middle layer (16 of 32), the same clustering, budgets and steering strength (0.4 times the mean activation norm). Nothing was tuned on Phi. The diagnosis replicates (Table 8). The probe's AUROC is 0.98–1.00 on all seven framings listed, while the shared threshold finds 0–88% of harmful prompts and the oracle threshold 82–90%. The hybrid rule with $k_{\max}$=30 finds 61–93% at 1.3% mean FPR (0.3% for the shared threshold), and over-refusal on plain benign requests moves from 23.9% to 24.8%. With the pre-registered cap of 12 clusters the gain is smaller (fiction 62%, indirect injection 23%, past tense 10%), as on Qwen. The failure on attack-heavy framings also replicates: under realistic traffic adaptive recall on persona and refusal suppression is 0%, and the hybrid floor recovers 63–65%. Base64 is again undetectable

		Benign-rich traffic			Realistic traffic		
Group	Framing	Shared	Adaptive	Hybrid	Shared	Adaptive	Hybrid
Seen	Direct	96	91	97	96	93	97
	Persona	88	85	85	87	4	58
	Ref. supp.	83	72	75	83	0	55
	Prefix inj.	93	88	87	93	0	81
Unseen	Fiction	5	19	19	4	9	9
	Indirect	16	31	29	16	28	26
	Base64	0	0	0	0	1	1
	Leetspeak	0	2	2	0	0	0
Fresh (pre-registered)	Past tense	0	15	14	0	7	6
	Academic	40	45	47	39	41	41
	Code	83	79	91	83	74	85
	Translation	96	94	94	95	93	93
	Payload split	0	90	89	0	0	0
	Benign FPR (mean)	0.6	1.5	1.4	0.7	1.2	1.3

(a) Pre-registered, $k_{\max}$=12, 20 windows

		Benign-rich traffic			Realistic traffic		
Group	Framing	Shared	Adaptive	Hybrid	Shared	Adaptive	Hybrid
Seen	Direct	96	76	95	96	70	94
	Persona	88	92	91	87	0	58
	Ref. supp.	83	96	96	83	0	56
	Prefix inj.	93	91	91	93	0	82
Unseen	Fiction	4	91	91	5	89	88
	Indirect	16	92	91	16	89	89
	Base64	0	1	1	0	1	1
	Leetspeak	0	3	3	0	0	0
Fresh (pre-registered)	Past tense	0	40	38	0	36	35
	Academic	40	84	83	40	80	79
	Code	83	82	92	83	83	90
	Translation	96	94	94	95	92	92
	Payload split	0	90	90	0	0	0
	Benign FPR (mean)	0.6	1.6	1.6	0.7	1.1	1.3

(b) Post hoc, $k_{\max}$=30, 10 windows

Table 5: Recall (%) on harmful test prompts of half B, mean over windows; last row: benign FPR averaged over the 13 framings. Thirteen framings share each traffic window. *Shared*: fixed threshold. *Adaptive*: cluster-wise robust z alone. *Hybrid*: either fires (each at half the budget).

Rule	Fiction	Indirect	Past tense	Academic	Payload split	FPR
Shared threshold	4	16	0	40	0	0.6
Global robust z (k=1)	1	5	0	30	0	0.4
Per-family z (oracle groups)	76	29	44	47	90	1.2
Per-cluster z (ours)	91	92	40	84	90	1.6

Table 6: Grouping ablation, benign-rich traffic, $k_{\max}$=30 (post hoc). Recall (%); FPR averaged over 13 framings.

Framing	No defense	Shared threshold	Adaptive (ours)	Hybrid (ours)
Direct	1 [0,2]	0 [0,1]	1 [0,2]	0 [0,1]
Persona	3 [2,5]	1 [0,3]	3 [2,5]	3 [2,5]
Ref. supp.	3 [1,3]	2 [1,3]	3 [1,3]	3 [1,3]
Prefix inj.	77 [74,82]	9 [6,13]	77 [74,82]	16 [13,21]
Fiction	89 [87,92]	88 [86,90]	36 [27,46]	37 [29,47]
Indirect	12 [9,17]	12 [9,17]	5 [2,9]	6 [2,9]
Leetspeak	0 [0,0]	0 [0,0]	0 [0,0]	0 [0,0]
Past tense	6 [4,7]	6 [4,7]	5 [3,6]	5 [3,6]
Academic	15 [13,17]	9 [8,12]	4 [2,8]	4 [2,8]
Code	45 [40,51]	14 [10,16]	7 [4,13]	6 [3,13]
Translation	51 [45,55]	3 [2,5]	5 [3,13]	5 [3,8]
Payload split	95 [93,97]	95 [93,97]	95 [93,97]	95 [93,97]
Over-refusal (plain benign)	60.3	60.3	60.6	60.6

Table 7: Judged ASR (%) on half B under realistic traffic, $k_{\max}$=30 (post hoc); mean and 95% interval over 10 windows. Base64 omitted (judge artefact, Section 3). Last row: over-refusal on plain benign requests.

(AUROC 0.39).

Generation results replicate for most framings but not for fiction. Judged ASR falls from 19% to 7% on past tense, 10% to 2% on academic, 15% to 8% on code, 13% to 3% on translate-then-answer, and 15% to 3% on payload splitting under benign-rich traffic. Under realistic traffic payload splitting is half harmful, its cluster is lost, and ASR stays at 15%. Fiction is the informative failure. The hybrid rule fires on 88% of harmful fiction prompts, yet ASR stays at 38%, because on Phi the refusal steer at this strength does not change fiction behaviour even when always on (36.9% undefended, 37.3% steered). The gate is right and the intervention is not. A fixed steering strength tuned on one model need not transfer, which is a limit of the intervention and not of the recalibration.

8 Discussion

By AUROC, activation gates generalise to new framings; at a deployed threshold they do not. This agrees with observations that probes keep their ranking but lose their operating point off-distribution (McKenzie et al., 2025; Kim and Lee, 2026; Azov et al., 2026), and it differs from settings where the information itself is missing (Schwarz, 2026), as for base64 on a small model. Defenses that remove framing from the representation, as ARETE attempted, address a change in separability, whereas here the change was in location.

The remedy's limits follow from one assumption, a bound on the harmful share of each cluster, and Section 5 makes both its necessity and its cost explicit. Recall is high when that share is about 5%, roughly halved at 20%, and mostly gone by 30% to 50%, so the assumption holds for dual-use framings such as fiction, retrieved documents, academic or code requests, which are also the hardest to block wholesale. It fails for framings that exist mainly to attack, and it can be forced to fail by an attacker who floods a cluster. Those cases need a fixed floor or a template-level filter. It also depends on cluster resolution: over-segmenting is cheap, because small clusters fall back to validation statistics, whereas under-segmenting silently merges framings.

9 Conclusion

Activation-gated jailbreak defenses fail on withheld framings mainly because their threshold stays where it was calibrated, while the model still represents the harm. Recalibrating per framing cluster from unlabeled traffic, with the original threshold as a floor, recovers much of the lost detection on two models, and much of the lost protection for

dual-use framings at almost no over-refusal cost. It cannot protect framings used mainly for attacks or flooded by an attacker, and it is only as effective as the intervention it triggers.

Limitations

We study two small models (1.5B and 3.8B parameters). The second was run with the configuration frozen from the first, but larger models were not tested. Larger models parse encodings such as base64, which would likely turn that framing from a capability failure into a calibration failure. The steering strength was fixed from the first model and does not transfer to every framing: on the second model it leaves fiction unchanged. Each framing uses one to three fixed templates, which makes clustering easier than for paraphrased or attacker-generated framings. The best-performing configuration ($k_{\max}$=30) was chosen after seeing pre-registered results on the first model, and the original withheld framings informed the method's design. The cleaner evidence is the fresh framings under the pre-registered configuration, and the second model, where $k_{\max}$=30 was applied without change. We tested one adaptive strategy, flooding a single cluster, and only on the first model. We did not test attackers who optimise against the gate (Bailey et al., 2026), spread a framing over many small clusters that fall back to validation statistics, or poison the window that sets the statistics. The location-scale model is validated against scores it was built from, so its agreement with measurements does not show that it predicts unseen traffic, and it describes the clustered rule only roughly. ASR is judged on 96-token responses by one model judge, and generation-level ASR is composed from undefended and always-on responses. The traffic mixes are simulated; real prevalence per framing is unknown.

Ethics Statement

This work studies defenses against misuse of language models. Harmful prompts come from public benchmarks, the attack templates are publicly documented, and generated responses are kept out of the released artefacts. We report failure modes, including attack-only framings and cluster under-segmentation, so that defenders do not over-trust activation gates.

References

Amr Alexandari, Anshul Kundaje, and Avanti Shrikumar. 2020. Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation. In *International Conference on Machine Learning (ICML)*.

Cem Anil, Esin Durmus, Nina Panickssery, Mrinank Sharma, et al. 2024. Many-shot jailbreaking. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Guy Azov, Ofer Rivlin, and Guy Shtar. 2026. SALLIE: Generation-free hidden-state detection of jailbreaks and prompt injections across text and vision. *arXiv preprint arXiv:2604.06247*.

Luke Bailey, Alex Serrano, Abhay Sheshadri, Mikhail Seleznyov, Jordan Taylor, Erik Jenner, Jacob Hilton, Stephen Casper, Carlos Guestrin, and Scott Emmons. 2026. Obfuscated activations bypass LLM latent-space defenses. In *International Conference on Learning Representations (ICLR)*.

Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwag, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramèr, Hamed Hassani, and Eric Wong. 2024. JailbreakBench: An open robustness benchmark for jailbreaking large language models. In *NeurIPS Datasets and Benchmarks Track*.

Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J. Pappas, and Eric Wong. 2025. Jailbreaking black box large language models in twenty queries. In *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*.

Justin Cui, Wei-Lin Chiang, Ion Stoica, and Cho-Jui Hsieh. 2025. OR-Bench: An over-refusal benchmark for large language models. In *International Conference on Machine Learning (ICML)*.

Shravan Doda. 2026. Before the last token: Diagnosing final-token safety probe failures. *arXiv preprint arXiv:2605.12726*.

Amirhossein Farzam, Majid Behabahani, Mani Malek, Yuriy Nevmyvaka, and Guillermo Sapiro. 2026. Hiding in plain text: Detecting concealed jailbreaks via activation disentanglement. *arXiv preprint arXiv:2602.19396*.

Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. 2023. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*.

Seungju Han, Kavel Rao, Allyson Ettinger, Liwei Jiang, Bill Yuchen Lin, Nathan Lambert, Yejin Choi, and Nouha Dziri. 2024. WildGuard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals of LLMs. In *NeurIPS Datasets and Benchmarks Track*.

Ursula Hébert-Johnson, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. 2018. Multicalibration: Calibration for the (computationally-identifiable) masses. In *International Conference on Machine Learning (ICML)*.

Xiaomeng Hu, Pin-Yu Chen, and Tsung-Yi Ho. 2024. Gradient cuff: Detecting jailbreak attacks on large language models by exploring refusal loss landscapes. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabsa. 2023. Llama guard: LLM-based input-output safeguard for human-AI conversations. *arXiv preprint arXiv:2312.06674*.

JaeSeong Kim and Suan Lee. 2026. Linearly readable, not robust: How linear jailbreak probes conflate benign register with adversarial intent. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 35071–35088, Budapest, Hungary.

Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehling, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar. 2025. Programming refusal with conditional activation steering. In *International Conference on Learning Representations (ICLR)*.

Christophe Leys, Christophe Ley, Olivier Klein, Philippe Bernard, and Laurent Licata. 2013. Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. *Journal of Experimental Social Psychology*, 49(4):764–766.

Xuan Li, Zhanke Zhou, Jianing Zhu, Jiangchao Yao, Tongliang Liu, and Bo Han. 2023. DeepInception: Hypnotize large language model to be jailbreaker. *arXiv preprint arXiv:2311.03191*.

Zachary C. Lipton, Yu-Xiang Wang, and Alexander J. Smola. 2018. Detecting and correcting for label shift with black box predictors. In *International Conference on Machine Learning (ICML)*.

Xiaogeng Liu, Nan Xu, Muhao Chen, and Chaowei Xiao. 2024. AutoDAN: Generating stealthy jailbreak prompts on aligned large language models. In *International Conference on Learning Representations (ICLR)*.

Mingyu Luo, Ming Deng, Zilang Qiu, Yiming Cheng, Ci Tao, Xue Tan, Sijin Sun, Yangfu Li, Ping Chen, Jun Dai, and Xiaoyan Sun. 2026. Measuring the wrong thing: Internal harmfulness scores anti-rank successful jailbreaks. *arXiv preprint arXiv:2608.09624*.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. 2024. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *International Conference on Machine Learning (ICML)*.

Alex McKenzie, Urja Pawar, Phil Blandfort, William Bankes, David Krueger, Ekdeep Singh Lubana, and Dmitrii Krasheninnikov. 2025. Detecting high-stakes interactions with activation probes. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, Blaine Nelson, Hyrum Anderson, Yaron Singer, and Amin Karbasi. 2024. Tree of attacks: Jailbreaking black-box LLMs automatically. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Lingrui Mei, Shenghua Liu, Yiwei Wang, Baolong Bi, Ruibin Yuan, and Xueqi Cheng. 2026. HiddenGuard: Fine-grained safe generation with specialized representation router. In *Proceedings of ACL (Long Papers)*.

Inkit Padhi et al. 2025. Granite guardian: Comprehensive LLM safeguarding. In *Proceedings of NAACL 2025 (Industry Track)*, pages 607–615.

Julien Piet, Xiao Huang, Dennis Jacob, Annabella Chow, Maha Alrashed, Geng Zhao, Zhanhao Hu, Chawin Sitawarin, Basel Alomair, and David Wagner. 2025. JailbreaksOverTime: Detecting jailbreak attacks under distribution shift. *arXiv preprint arXiv:2504.19440*.

Aleksandr Podkopaev and Aaditya Ramdas. 2021. Distribution-free uncertainty quantification for classification under label shift. In *Conference on Uncertainty in Artificial Intelligence (UAI)*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Qwen Team, Haiquan Zhao, et al. 2025. Qwen3Guard technical report. *arXiv preprint arXiv:2510.14276*.

Paul Röttger, Hannah Rose Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. 2024. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of NAACL-HLT*.

Peter J. Rousseeuw and Christophe Croux. 1993. Alternatives to the median absolute deviation. *Journal of the American Statistical Association*, 88(424):1273–1283.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. 2002. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41.

Dominik Schwarz. 2026. The entanglement wall: Activation-space probes as risk detectors, not context adjudicators. *arXiv preprint arXiv:2607.13075*.

Mrinank Sharma, Meg Tong, Jesse Mu, Jerry Wei, et al. 2025. Constitutional classifiers: Defending against universal jailbreaks across thousands of hours of red teaming. *arXiv preprint arXiv:2501.18837*.

Xinyue Shen, Zeyuan Chen, Michael Backes, Yun Shen, and Yang Zhang. 2024. "Do Anything Now": Characterizing and evaluating in-the-wild jailbreak prompts on large language models. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*.

Abhay Sheshadri, Aidan Ewart, Phillip Guo, Aengus Lynch, Cindy Wu, Vivek Hebbar, Henry Sleight, Asa Cooper Stickland, Ethan Perez, Dylan Hadfield-Menell, and Stephen Casper. 2024. Latent adversarial training improves robustness to persistent harmful behaviors in LLMs. *arXiv preprint arXiv:2407.15549*.

Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter Abbeel, Justin Svegliato, Scott Emmons, Olivia Watkins, and Sam Toyer. 2024. A StrongREJECT for empty jailbreaks. In *NeurIPS Datasets and Benchmarks Track*.

Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. 2019. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. 2021. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations (ICLR)*.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. 2023. Jailbroken: How does LLM safety training fail? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Yueqi Xie, Minghong Fang, Renjie Pi, and Neil Gong. 2024. GradSafe: Detecting jailbreak prompts for LLMs via safety-critical gradient analysis. In *Proceedings of ACL (Long Papers)*.

Youliang Yuan, Wenxiang Jiao, Wenxuan Wang, Jen-tse Huang, Pinjia He, Shuming Shi, and Zhaopeng Tu. 2024. GPT-4 is too smart to be safe: Stealthy chat with LLMs via cipher. In *International Conference on Learning Representations (ICLR)*.

Wenjun Zeng, Yuchi Liu, Ryan Mullins, Ludovic Peran, et al. 2024. ShieldGemma: Generative AI content moderation based on Gemma. *arXiv preprint arXiv:2407.21772*.

Shenyi Zhang, Yuchen Zhai, Keyan Guo, Hongxin Hu, Shengnan Guo, Zheng Fang, Lingchen Zhao, Chao Shen, Cong Wang, and Qian Wang. 2025. JBShield: Defending large language models from jailbreak attacks through activated concept analysis and manipulation. In *34th USENIX Security Symposium*, pages 8215–8234.

Jiachen Zhao, Jing Huang, Zhengxuan Wu, David Bau, and Weiyan Shi. 2025. LLMs encode harmfulness and refusal separately. *arXiv preprint arXiv:2507.11878*.

Albert Ziegler and Paweł Czyż. 2019. Unsupervised recalibration. *arXiv preprint arXiv:1908.09157*.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, and 2 others. 2023a. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*.

Andy Zou, Long Phan, Justin Wang, Derek Duenas, Maxwell Lin, Maksym Andriushchenko, Rowan Wang, Zico Kolter, Matt Fredrikson, and Dan Hendrycks. 2024. Improving alignment and robustness with circuit breakers. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. 2023b. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*.

A ARETE Geometry

ARETE standardises activations and maps them through two learned linear heads onto unit hyperspheres for intent and context. It is trained with (i) an invariance loss pulling the intent coordinates of a wrapped prompt towards those of its plain version, (ii) a framing-classification loss on the context sphere, and (iii) a cap loss placing harmful intents inside, and benign intents outside, eight unsupervised spherical caps. The gate score is the maximum cap margin. Steering uses the refusal direction projected onto the orthogonal complement of a rank-16 framing subspace, estimated by SVD of displacements $h_{\text{wrapped}} - h_{\text{plain}}$ on training behaviours.

B Additional Results

Second model. Table 8 gives the Phi-3.5-mini replication discussed in Section 7.

		Recall (%)				Judged ASR (%)		
Framing	AUROC	Shared	Oracle	Hyb. 12	Hyb. 30	None	Shared	Hyb. 30
Fiction	1.00	47	90	62	88	38	46	38
Indirect	0.99	21	90	23	93	3	2	1
Past tense	0.98	0	83	10	61	19	19	7
Academic	1.00	68	89	69	91	10	3	2
Code	0.99	74	82	79	90	15	10	8
Translation	1.00	88	89	89	92	13	5	3
Payload split	1.00	20	90	68	93	15	14	3
Benign FPR (mean)		0.3		1.0	1.3			

Table 8: Replication on Phi-3.5-mini-instruct with the configuration frozen from Qwen2.5-1.5B (user-mean probe, 1% budget, no tuning), benign-rich traffic, 10 windows. AUROC and *Oracle* recall are for the user-mean probe on the full test split; *Shared*, *Hyb. 12* and *Hyb. 30* are recall of the shared threshold and of the hybrid rule with $k_{\max}$=12 and 30. Judged ASR is composed as in Section 7.

	harmful prevalence π				
d	0.05	0.10	0.20	0.30	0.50
2	23.9	17.8	7.8	2.4	0.2
3	60.1	49.6	23.3	3.6	0.0
4	89.5	83.5	58.0	12.7	0.0
5	98.8	97.6	88.4	42.3	0.0

Table 9: Recall (%) of the oracle-grouped robust-z rule under the Gaussian model, by separation d and prevalence π.

		adaptive, π =				hybrid, π =			
framing	shared	0.05	0.2	0.3	0.5	0.05	0.2	0.3	0.5
fiction	4.9	86.9	58.6	17.4	0.0	85.6	56.5	15.4	0.8
code	83.3	84.1	49.0	18.1	0.0	91.2	78.7	69.2	69.0
academic	40.4	85.5	42.0	5.9	0.0	84.1	48.7	30.0	24.6
indirect	16.9	91.5	67.4	29.8	0.0	90.9	65.4	28.3	2.7
past tense	0.6	43.1	13.3	5.6	0.0	41.7	11.4	4.7	0.0

Table 10: Recall (%) of the adaptive and hybrid rules on one framing as its harmful prevalence π grows.

Gaussian breakdown curve. Table 9 gives the recall of the oracle-grouped robust-z rule under the Gaussian model of Proposition 3, as a function of separation d and harmful prevalence π, at $z^\star = 2.52$, the value fitted on Qwen2.5-1.5B.

Flooding. Table 10 reports recall (%) on one target framing as its harmful share π is raised while all other framings keep their realistic prevalence, on Qwen2.5-1.5B with $k_{\max}$=30 and 10 windows. The shared column is the shared threshold at full budget, averaged over the same windows. All five framings we flooded are shown; the main text discusses fiction and code.

Pre-registered configuration at generation level. Table 11 repeats Table 7 with $k_{\max}$=12.

Framing	No defense	Shared threshold	Adaptive (ours)	Hybrid (ours)
Direct	1 [0,2]	0 [0,1]	1 [0,2]	0 [0,1]
Persona	4 [2,6]	1 [0,3]	4 [2,6]	4 [2,5]
Ref. supp.	3 [1,4]	2 [1,3]	3 [1,4]	3 [1,4]
Prefix inj.	76 [70,82]	10 [6,13]	76 [70,82]	17 [13,21]
Fiction	89 [85,92]	87 [83,92]	85 [67,92]	86 [68,93]
Indirect	13 [10,18]	13 [10,18]	13 [10,18]	13 [10,18]
Leetspeak	0 [0,0]	0 [0,0]	0 [0,0]	0 [0,0]
Past tense	6 [4,8]	6 [4,8]	6 [4,8]	6 [4,8]
Academic	15 [11,18]	9 [5,12]	9 [2,16]	9 [3,14]
Code	46 [40,53]	14 [9,18]	11 [2,27]	12 [3,26]
Translation	52 [46,56]	3 [2,5]	5 [0,11]	4 [1,8]
Payload split	95 [93,97]	95 [93,97]	95 [93,97]	95 [93,97]
Over-refusal (plain benign)	59.6	59.6	60.6	60.5

Table 11: As Table 7, pre-registered $k_{\max}$=12, 20 windows.

Prevalence. Table 12 uses the eight original framings, uniform prevalence π and 20 draws.

GCG. Suffixes were optimised for 20 test behaviours against the undefended model (80 steps, 128 candidates per step, top-128 tokens, 20-token suffix). They reach 25% judged ASR undefended. The final-token linear probe lowers this to 10%, the multi-view union to 15%, and ARETE to 22%. With 20 behaviours these numbers are indicative only.

Longer responses. The main tables judge responses truncated at 96 tokens, which could mostly measure the onset of compliance. We regenerated the undefended and always-on responses of seven framings for all 241 harmful test prompts at 256 tokens and re-judged them. Attack success barely moves. Without defense it is 88.8% (96 tokens) versus 89.2% (256) on fiction, 75.9% versus 74.7% on prefix injection, and 95.4% versus 94.2% on payload splitting. With the always-on steer it is 28.6% versus 29.9%, 4.6% versus 5.0%, and 0.4%

		Seen framings (TPR)				Unseen framings (TPR)				
π	Rule	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak	FPR
1%	Shared	98	92	80	92	5	20	0	0	0.5
	Adaptive	88	95	100	92	85	70	0	5	1.8
	Hybrid	98	95	100	92	85	68	0	5	1.7
5%	Shared	95	86	85	91	7	17	0	0	0.5
	Adaptive	80	90	94	88	87	65	0	4	1.6
	Hybrid	93	88	94	87	86	64	0	2	1.5
10%	Shared	96	89	83	92	4	17	0	0	0.5
	Adaptive	73	87	86	82	78	56	0	1	1.3
	Hybrid	94	86	84	82	76	54	0	1	1.3
20%	Shared	97	87	83	92	5	17	0	0	0.5
	Adaptive	28	50	61	56	52	36	0	1	0.8
	Hybrid	95	59	76	81	49	35	0	0	1.0

Table 12: Eight original framings, uniform harmful prevalence π, $k_{\max}$=12; recall (%) and benign FPR (%) averaged over framings. At π=35% (not shown) adaptive recall is at most 5% on every framing.

versus 0.8%. The gap between steered and undefended responses, which drives every composed ASR in the main text, therefore does not depend on the truncation length.